Collaborative Multivariate Time Series Forecasting via Variable-Tailored Inter-Temporal Graph and Adaptive-Smooth Frequency Fusion

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Abstract

Frequency domain learning and accurate multivariate dependencies are crucial for driving multivariate time series forecasting applications in real world. However, the existing progress remains limited. First, the multivariate variables can be divided into "Multi-Attribute" and "Multi-Entity" types, which necessitate considerations for tailored correlation modeling and dynamic dependencies capturing. Second, the inherent non-stationary of time series conflicts with the stationary assumptions of frequency analysis, and the temporal globality of Fourier basis function tends to neglect local information. To address these challenges, we propose the Variable-Tailored Inter-Temporal Graph and Adaptive-Smooth Frequency Fusion Network (VTITG-ASFFN), which first adaptively stabilizes time series through mask learning and realizes local-global collaborative learning by frequency components mining and fusion, then a notable innovation is a tailored inter-temporal graph for "Multi-Attribute" and "Multi-Entity" correlation scenarios, which effectively interacts with input series via Variable-Tailored Adaptive Graph and Channels-Time Graph, learning "dynamic spatial-temporal dependencies" in temporal context, enabling high-fidelity evolution of "Multi-Attribute" and dynamic understanding of correlations among "Multi-Entity". Evaluations on 8 real-world datasets demonstrate the superiority of VTITG-ASFFN in forecasting, efficiency and universality over SoTA benchmarks. The code is available at Github/VTITG-ASFFN.

Keywords: Multivariate time series forecasting; Multivariate relationships modeling; Frequency domain analysis; Sequence learning

1 Introduction

Multivariate Time Series Forecasting (MTSF) is a fundamental research area in the big data era (Torres et al. 2018). It finds extensive applications in various domains, such as electricity consumption, solar power generation, software system load evaluation, traffic forecasting, and meteorological prediction (Papastefanopoulos et al. 2023; Sun et al. 2022). Rethinking MTSF from the advanced deep learning perspective, Temporal Convolutional Networks (TCN) (Wu et al. 2022; Luo et al. 2024) are employed to capture temporal patterns by expanding receptive fields; For more accurate long-term prediction, Transformer-based methods (Wu et al. 2021; Zhou et al. 2022; Nie et al. 2023; Liu et al. 2024) use the stacked encoder-decoder architecture and point-wise attention mechanism to delve into the long-term dependencies, facilitating the emergence of LLM (Jin et al. 2024b) for time series; Adaptive graphs (Wu et al. 2019) excel at capturing spatial dependencies in spatio-temporal data, which also has been extended to model variable relationships in MTSF (Huang et al. 2023; Cai et al. 2024). Furthermore, models based on simple deep learning components are gaining attention. Many strategies (decomposition, sampling, mixing) for modeling periodic and variable relationships are integrated on MLP-based structure (Zeng et al. 2023; Zhang et al. 2022; Wang et al. 2024). Currently, a more promising direction focus on effectively leveraging the frequency domain information (Zhou et al. 2022; Wu et al. 2022; Yi et al. 2024a; Wu et al. 2021) to achieve more accurate multivariate time series forecasting. In the frequency domain, time series are decomposed into frequency components, rendering periodic information more explicit for long-term planning and early warning (Jin et al. 2024a). Although the frequency domain provides a new perspective, several essential MTSF issues still remained and have become pressing challenges, involving the multivariate natures, dynamic spatial-temporal patterns and the applicable conditions for frequency domain analysis. We have elaborated on them in detail:

Challenge I: **Variable Type Difference.** The multivariate natures are rarely noticed in previous studies. Due to the conceptual differences in the variables types, we define "Multi-Attribute Variables (MAV)" and "Multi-Entity Variables (MEV)" forecasting tasks. Multi-attributes reflect the physical properties (e.g., atmospheric pressure and temperature), while Multi-entities indicate records of different entities for the same observation (e.g., traffic sensors). In addition, we have observed the auto-correlation variation in Fig. 1, which demonstrates that: (1) Auto-correlation among multivariate attributes has significant differences, reflecting the temporal evolution of

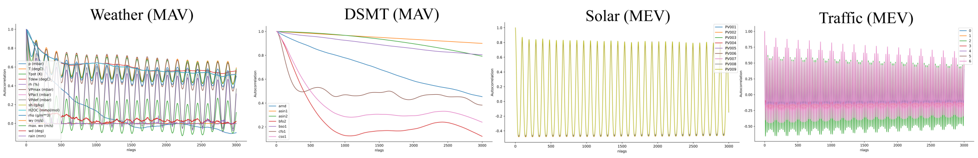

Fig. 1 Temporal auto-correlation variations in "MAV" and "MEV" scenarios.

the physical system; (2) In contrast, the auto-correlation of multi-entities is more similar in the long term. These differences illustrate that "MAV" models should account for attribute-specific dynamics, while "MEV" models can leverage shared patterns.

Challenge II: **Dynamic Spatial-Temporal Dependencies Capture.** Self-adaptive graphs have the ability to capture multivariate relationships without prior knowledge. However, they are limited to learn the static graph representations, which fail to establish connections with temporal evolution. In other words, they overlook the critical aspect of "dynamic spatial-temporal dependencies" (Yi et al. 2024b). We have observed the daily electricity usage data in a specific real-world region in Fig. 3 (d) top. It was found that while the neighboring households tend to follow similar patterns (Shao et al. 2022), which is also based on specific time periods. Therefore, multivariate dependencies are inherently diverse in temporal context.

Challenge III: **Ignoring Localization and Stationary.** Frequency domain analysis is inclined to capture global periodic patterns, yet it may neglect local details (Zhuang et al. 2024). This issue arises from the Fourier basis function, due to its time-axis globality, transient changes at specific time points are spread across the entire frequency domain (Fan et al. 2024), thus obscuring temporal localization. Additionally, many real-world time series are non-stationary (Casals et al. 2000), which may lead to misleading spectral patterns, complicating stationary frequency analysis.

For these challenges, we primarily offer contributions for jointly improving frequency domain learning and tailored dynamic graph modeling: **(1) Tailored Dynamic Graph Modeling.** We propose a two-stage graph modeling framework. In the first stage, from the observations of MTS, we establish static correlations by self-adaptive graph tailored to "MAV" and "MEV". In the second stage, we dynamize multivariate correlations and transfer them into temporal context by channels-time graph to capture "dynamic spatial-temporal dependencies". **(2) Local Frequency Learning and Global Fusion.** We conduct channel-independent embedding learning paradigm (Nie et al. 2023; Zeng et al. 2023), which will focus on mining latent patterns for discrete frequency components (i.e., refining granularity from coarse to fine), thus alleviating the lack of local information in frequency-domainized time series and synergizing modal complementarity (amplitude and phase) for global fusion (Loweimi et al. 2023). **(3) Adaptively Smoothing.** To ensure smooth (i.e., stable) time series before frequency domain learning, we propose an adaptive masking mechanism. This idea is inspired by the Short-Time Fourier Transform (STFT) (Pei and Huang 2012), where the "window function" smoothes the signal and reduces noise. The adaptive mask acts as a dynamic "window function", enabling targeted smoothing to retain robust statistical properties in spectral analysis and mitigate the impact of non-stationarity.

In this paper, we introduce a collaborative multivariate time series forecasting model, namely Variable-Tailored Inter-Temporal Graph and Adaptive-Smooth Frequency Fusion Network (VTITG-ASFFN). Our main contributions are as follows:

- For temporal learning, to address the limitations of frequency analysis in neglecting local information and their reliance on the assumption of sequence stationary, we designed the Adaptive-Smooth Frequency Fusion MLP, realizing collaborative local-global learning and targeted smoothing guidance for diverse time series; To comprehensively integrate, we also introduce the Frequency Dilated Group Convolution, which facilitates multi-source signal fusion and primary temporal patterns mining.
- For multivariate correlations, we design Variable-Tailored Inter-Temporal Graph to capture "dynamic spatial-temporal dependencies", which first provides tailored adaptive graphs for "MAV" and "MEV", then uses channels-time graph to incorporate these correlations into the temporal context for diverse spatial-temporal patterns.
- Comprehensive evaluations conducted on 8 real-world datasets confirms that VTITG-ASFFN achieves superior performance across all SoTA benchmarks.

2 Related Work

2.1 Multivariate Time Series Forecasting

Early time series forecasting primarily relied on traditional statistical models and classical machine learning approaches. The ARIMA model (Shumway et al. 2017) integrates three components: autoregression (AR), differencing (I), and moving average (MA), generating forecasts based on historical time-window data. Gradient boosted regression trees (GBRT) (Chen and Guestrin 2016; Han et al. 2024), by ensembling multiple weak learners with temporal feature engineering, demonstrated competitive performance in specific scenarios. However, such models face inherent limitations in efficiently parallelizing multivariate time series forecasting, and their dominance has been challenged by the advancement of deep learning techniques. We focus on the advanced MTSF methods including MLP, TCN, Transformer, LLM and Graph based on deep learning. For MLP-based, DLinear (Zeng et al. 2023) utilizes a simple linear layer to independently predict the seasonal and trend components of the variables; TimeMixer (Wang et al. 2024) breaks down mixed time variations into multiple components (de-entanglement) to highlight the intrinsic properties of the time series. For TCN-based, ModernTCN (Luo et al. 2024) jointly designs depth-wise and grouped point-wise convolution to learn cross-variable and temporal dependencies. For Transformer-based, PatchTST (Nie et al. 2023) proposes patch modeling with channel independent approach and Transformer; iTransformer (Liu et al. 2024) utilizes inverted dimensional attention and encode variable as token for representations learning. For LLM-based, Time-LLM (Jin et al. 2024b) recodes time series through cross-modal attention and uses domain knowledge to guide LLM inference. However, they mainly focus on temporal patterns. From graph structure learning perspective, although CrossGNN and MSGNet (Huang et al. 2023; Cai et al. 2024) establish adaptive graph across different time scales, they heavily rely on manually defined scale extraction, making it challenging to capture "dynamic spatial-temporal dependencies". What's more, none of them try to take into account multivariate types (***Challenge*** *I*).

2.2 Self Adaptive Graph Modeling

Adaptive graph is used to capture spatial dependencies in traffic flow. Graph WaveNet (Wu et al. 2019) initializes the source and target embedding for each node and builds asymmetric matrix to represent the directed relationships. AGCRN (Bai et al. 2020) provides only a single embedding for each node and computes the similarity. MTGNN (Wu et al. 2020) uses the argtopk algorithm to dynamically select the most relevant k-node pairs to reduce the computational complexity. DMSTGCN (Han et al. 2021) constructs dynamic adaptive matrix for multifaceted inter-node relationship capture and fusion. MegaCRN (Jiang et al. 2023) learns spatio-temporal heterogeneity with Meta-Graphs. However, none of them have attempted to transfer the multivariate correlations of adaptive graph into temporal context for further temporal correlations mining, failing to capture diverse spatio-temporal dependencies (***Challenge*** II).

2.3 Frequency Domainization

Existing works focus on Fourier transform to appropriately represent time series. Autoformer (Wu et al. 2021) designs an auto-correlation mechanism based on the Fourier transform to capture sequence periodicity for correlation discovery and representation aggregation at the subsequence level. Fedformer (Zhou et al. 2022) combines seasonal and trend decomposition, Fourier analysis and Transformer to better capture global properties of time series. TimesNet (Wu et al. 2022) inspired by multi-periodicity to capture intra- and inter-periodic variations in frequency domain. FilterNet (Yi et al. 2024a) selectively passes components by learnable frequency filters. However, they seldom consider "adaptively" pre-smoothing the time series, meanwhile lacking the local-global collaborative learning for frequency components (***Challenge*** III).

3 Preliminary

(1) Problem Definition. The Multivariate Time Series (MTS) Forecasting refers to predict future period based on historical time series in time windows form (i.e., look-back window), that is $\{X_{t+S_h}^{\{1,...,c\}}, ..., X_{t+S_h+S_f-1}^{\{1,...,c\}}\} = f_\theta(X_t^{\{1,...,c\}}, ..., X_{t+S_h-1}^{\{1,...,c\}})$. Where S_h and S_f are the time step length of historical time series and future time series. c is the number of MTS variables. $f_\theta(\cdot)$ denotes MTSF models with parameter set $\{\theta\}$.
(2) Self-Adaptive Graph Structure. For multivariates, establishing embeddings $C_{\text{emb}} \in \mathbb{R}^{c\times d}$. The adaptive graph $G_{\text{adp}} \in \mathbb{R}^{c\times c}$ is formed by applying transposed multiplication to C_{emb}. The optimal learning process of G_{adp} can be derived from MTS observation set $\{\mathcal{O}\}$: $P(G_{\text{adp}} \in \{\mathcal{A}\}) = \sum_{X\in\{\mathcal{O}\}} P(X)P(G_{\text{adp}}|X)$, here $\{\mathcal{A}\}$ is optimal set for all learnable parameters, $P(\cdot)$ indicates probability of existence.
(3) Channels-Independent MLP. Unlike Channels-Dependent (CD) strategy, Channels-Independent MLP (CI-MLP) performs independent channel weight and bias W^{CI}, $\mathrm{bias}^{\mathrm{CI}}$ for MTS, disregarding cross-variable correlations: $Y^{\mathrm{CI}} = \sum_i X_{(b,c,i)}\mathrm{W}^{\mathrm{CI}}_{(i,1,c,o)} + \mathrm{bias}^{\mathrm{CI}}_{(1,c,o)}$. Where b, c, i, o denote the batch size, the number of variables, the length of the input and output MTS respectively. $Y^{CI} \in \mathbb{R}^{b\times c\times o}$.
(4) Globality in Discrete Fourier Transform. When signal $x[t]$ has transient change $\delta(t)$ in $t \in [t_1, t_2]$. Since frequency component $X[k] = \sum_{t=0}^{T-1} x[t]e^{-i2\pi k\frac{t}{T}}$ and

Fourier basis function $e^{-i2\pi k\frac{t}{T}}$ traverses the entire time domain, each $X[k]$ in $k \in [1, \lfloor\frac{T}{2}\rfloor + 1]$ contains the global information, leading to localization loss.

(5) Error-Driven Adaptive Smoothing. Suppose that for a frequency-domain based model $\mathrm{MF}(\cdot)$, a learnable mask $M \in \mathbb{R}^{c\times(\lfloor\frac{t}{2}\rfloor+1)}$ is introduced for pre-smoothing the MTS while minimizing the prediction error through $L1$ loss function, that is

$$\min \mathcal{L}(\mathrm{MF}, M) = \mathbb{E}_t\left[\|\mathrm{MF}(X(t) + Z(t)) - Y_{\mathrm{true}}(t)\|_1\right] \tag{1}$$

here $Z(t) = \mathcal{F}^{-1}\{M \odot \mathcal{F}\{X(t)\}\}$ indicates decoupled smooth information from forward and inverse Fourier transform $\mathcal{F}\{\cdot\}/\mathcal{F}^{-1}\{\cdot\}$, thus M can be updated by back-propagation $\frac{\partial\mathcal{L}(\mathrm{MF},M)}{\partial M} = \frac{\partial\mathcal{L}(\mathrm{MF},M)}{\partial\mathrm{MF}(\cdot)} \cdot \frac{\partial\mathrm{MF}(\cdot)}{\partial Z(t)} \cdot \frac{\partial Z(t)}{\partial M}$. Note that strictly smooth MTS are rare in practice (Casals et al. 2000), therefore we mainly propose weak smoothness.

4 Methodology

In this section, we will delve into our proposed framework VTITG-ASFFN and its core component modules. The VTITG-ASFFN architecture in Fig. 2, which consists of History-Embedding *(a)*, Prediction *(b)* and Future-Embedding *(c)*. We introduce Adaptive-Smooth Frequency Fusion MLP (ASFF-MLP), Variable-Tailored Inter-Temporal Graph (VT-ITGraph), and Frequency Dilated Group Convolution (FDG-Conv) to create stackable Frequency Domain Multivariate Block (FDMB), fully leveraging them for end-to-end embedding learning. Furthermore, we use CI-MLP for predicting future time series based on historical data. The VTITG-ASFFN operates through process: ***History-Embedding*** → ***Prediction*** → ***Future-Embedding***.

4.1 Adaptive-Smooth Frequency Fusion MLP

Adaptive-Masking. Before mapping the MTS into the frequency domain, we first define a learnable meta-embedding $M_{1,c,\lfloor\frac{t}{2}\rfloor+1} \in \mathbb{R}^{b\times c\times(\lfloor\frac{t}{2}\rfloor+1)}$. In analogy to the Short-Time Fourier Transform $\int_{-\infty}^{\infty} X(\tau)M^*(t-\tau)e^{-jwt}d\tau$ (τ means time point), $M_{1,c,\lfloor\frac{e}{2}\rfloor+1}$ can be viewed as an adaptively learned "window function" for frequency filtering:

$$P(M \in \{\mathcal{A}\}) = \sum\nolimits_{X\in\{\mathcal{O}\}} P(X, \mathrm{FFT}\{X\})P(M|X, \mathrm{FFT}\{X\}). \tag{2}$$

Through Fast Fourier Transform (FFT), the MTS are converted into frequency components, denoted as $F_{b,c,\lfloor\frac{t}{2}\rfloor+1}$. Subsequently, we leverage the Hadamard product $\odot$ to multiply $M_{1,c,\lfloor\frac{t}{2}\rfloor+1}$ with $F_{b,c,\lfloor\frac{t}{2}\rfloor+1}$ to get decoupled $F_{\mathrm{masked}} \in \mathbb{R}^{b\times c\times(\lfloor\frac{t}{2}\rfloor+1)}$:

$$F_{\mathrm{masked}} = M_{1,c,\lfloor\frac{t}{2}\rfloor+1} \odot F_{b,c,\lfloor\frac{t}{2}\rfloor+1}. \tag{3}$$

The extracted F_{masked} will adaptively delve into the smooth information via error-driven gradient descent, as it is considered as prerequisite for frequency learning during training. Meanwhile, dynamism manifests in the capability of F_{masked} to autonomously update with diverse MTS input. Ultimately, it is inversely transformed back into the

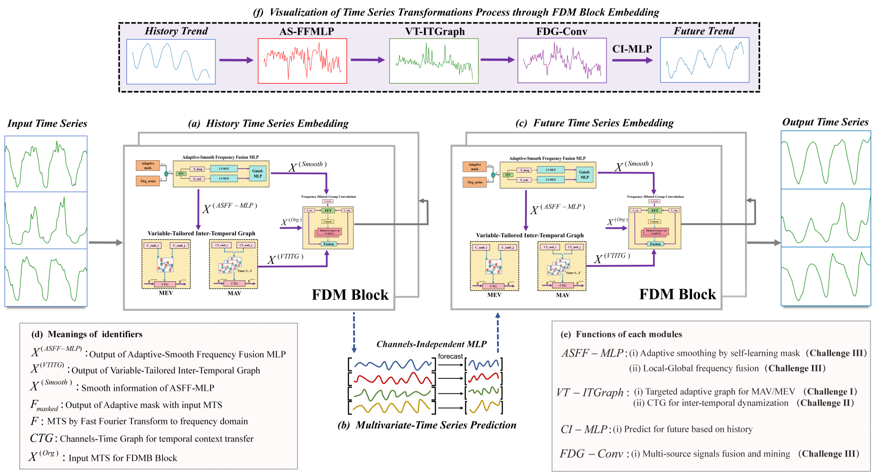

Fig. 2 The VTITG-ASFFN architecture for historical and future MTS embedding learning.

time domain and appended to original MTS $X^{(\text{Org})}$, mitigating the non-stationary:

$$X^{(S)} = X^{(\text{Org})} + \frac{1}{2\pi}\int_{-\infty}^{+\infty} F_{\text{masked}}(e^{jw})e^{jwt}dw \tag{4}$$

here $X^{(S)} \in \mathbb{R}^{b\times c\times t}$ denotes the smoothed MTS, which fulfills the prerequisite of stationary implicitly and lays the foundation for collaborative local-global patterns learning in frequency domain. Notably, the $\mathcal{F}^{-1}\{F_{\text{masked}}\}$ is remembered by $X^{(\text{Smooth})}$ for the smoothing of FDG-Conv in 4.3. w, f are angular and frequency factors.

To analyze the adaptivity from the gradient perspective, the error-driven back-propagation term for the k-th frequency component can be simplified to $\frac{\partial X^{(S)}(t)}{\partial M_k} = \mathcal{F}\{X^{(\text{Org})}\}_k \cdot \frac{1}{T}e^{i2\pi k\frac{t}{T}}$. Furthermore, since $\frac{\partial \mathcal{L}}{\partial M_k} \propto \frac{\partial X^{(S)}(t)}{\partial M_k}$, this implies that if a frequency component k in the input signal consistently leads to prediction error, a steep gradient will be generated causing M_k to approach zero. Conversely, if a frequency component is crucial for accurate prediction, the gradient will reinforce its magnitude in M to attenuate noise and amplify signal-bearing components selectively.

Local-Global Learning in Frequency Domain. Stationarized MTS $X^{(S)}$ are easy to analyze in the frequency domain for robust statistical features (Yi et al. 2024a). The Fast Fourier Transform method is applied to extract the spectral features of $X^{(S)}$, we regared them as the real part $X^{(S)}_{\text{real}}$ and the imaginary part $X^{(S)}_{\text{imag}} \in \mathbb{R}^{b\times c\times(\lfloor\frac{t}{2}\rfloor+1)}$:

$$X^{(S)}_{\text{real}}, X^{(S)}_{\text{imag}} = \int_{-\infty}^{+\infty} X^{(S)}_{(t)} cos(2\pi f t)dt, j\int_{-\infty}^{+\infty} X^{(S)}_{(t)} sin(2\pi f t)dt. \tag{5}$$

Specifically, $X^{(S)}_{\text{real}}$ and $X^{(S)}_{\text{imag}}$ exhibit the amplitude and phase distributions of different frequency components, and together they define the waveform and energy distribution of the signal. CI-MLP is employed to individually project the $X^{(S)}_{\text{real}}/X^{(S)}_{\text{imag}}$, performing

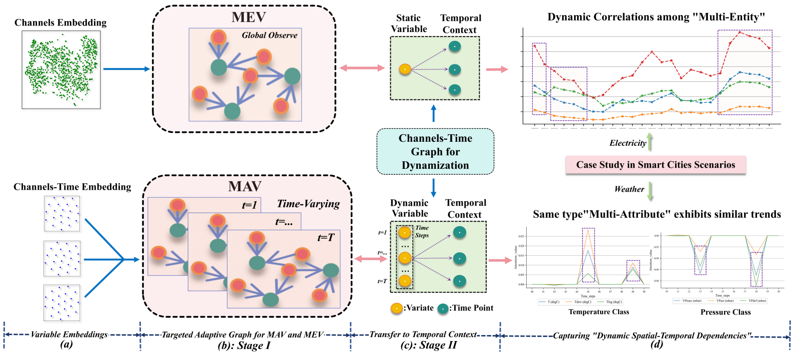

Fig. 3 Detailed layout of Variable-Tailored Inter-Temporal Graph. Top: VT-ITGraph for MEV task; Bottom: VT-ITGraph for MAV task. The VT-ITGraph consists of a tailored Adaptive Graph learner $G_{\text{adp}}^{\text{mav/mev}}$ and a Channels-Time Graph learner G_{CT}.

fine granular learning for latent local patterns. These projections are then summed:

$$X^{(\text{ASFF-MLP})} = \text{Gated-MLP}(\text{CI-MLP}(X_{\text{real}}^{(S)}) + \text{CI-MLP}(X_{\text{imag}}^{(S)})). \tag{6}$$

Where $X^{(\text{ASFF-MLP})}$ realizes local-global fusion, while in the dynamic fusion of real and imaginary components, we harness a Gated-MLP$(\cdot)$ (consists by Dropout, Tanh, MLP) to dynamically adjust the fusion weights, mitigating modal inconsistencies in amplitude and phase, and adjusting the temporal dimension $\mathbb{R}^{b\times c\times(\lfloor \frac{t}{2} \rfloor+1)} \rightarrow \mathbb{R}^{b\times c\times t}$ to keep the embedding dimensionality invariance of local-global fusion representation.

4.2 Variable-Tailored Inter-Temporal Graph

Existing studies (Bai et al. 2020; Wu et al. 2020; Han et al. 2021; Jiang et al. 2023) lack the dynamic utilization of adaptive graph within coherent temporal context. Therefore, VT-ITGraph is precisely designed for extending multivariate correlations to temporal context through self-learning the MTS observation, indirectly dynamizing static representation of adaptive graph. Formally, Inter-Temporal Graph can be likened to the attention form: $Q(X) \times K(G_{\text{adp}}, G_{\text{CT}}) \times V(X)$. Let input MTS be Q and V, $K(G_{\text{adp}}, G_{\text{CT}})$ means the graph learning of "dynamic spatial-temporal dependencies":

$$P(K(G_{\text{adp}}, G_{\text{CT}}) \in \{\mathcal{A}\}) = \sum\nolimits_{X\in\{\mathcal{O}\}} P(X)P(G_{\text{adp}}|X)P(G_{\text{CT}}|G_{\text{adp}}, X) \tag{7}$$

here $K(G_{\text{adp}}, G_{\text{CT}})$ like a dynamic meta-knowledge base, and its optimal learning process can be expressed by the conditional probability and chain rule. In Fig. 3, we can observe that the primary distinctions between VT-ITGraph in modeling MAV and MEV tasks lie in the generation methods of adaptive graphs, which stem from the differing semantics conveyed by these graphs. For MAV tasks, we propose a time-varying adaptive graph on the Fig. 3 bottom, as the temporal correlations between attribute variables varies significantly; For MEV tasks, we use a global self-adaptive graph on the Fig. 3 top because the temporal correlations of entity variables is similar overall (refer to ***Challenge*** *I* in 1). Finally they are all aggregated by the Channels-Time Graph. Therefore, we introduce VT-ITGraph via two-stage graph construction.

Adaptive Graph for MAV and MEV. Our modeling for "Multi-Attribute" and "Multi-Entity" variables is based on different underlying data generation assumptions. For MAV we assume that each variable is controlled by its own distinct latent process $\mathcal{Z}c$ (e.g., meteorological attributes) and its joint probability is factorized as $\prod_c P(X_c|\mathcal{Z}c, \theta_c) \cdot \Psi(X_1, \ldots, X_c; t)$. Where we employ a time-varying adaptive graph $G^{\mathrm{mav}}_{\mathrm{adp}} \in \mathbb{R}^{c\times c\times t}$ to explicitly model this dynamic interaction term $\Psi(X_1, \ldots, X_c; t)$. For MEV we assume that all variables are driven by a shared latent factor $\mathcal{Z}$ (e.g., regional traffic systems). Its joint probability distribution can be expressed as $\int P(\mathcal{Z}) \prod_c P(X_c|\mathcal{Z}, \theta_c) d\mathcal{Z}$. Based on this, we use a global self-adaptive graph $G^{\mathrm{mev}}_{\mathrm{adp}} \in \mathbb{R}^{c\times c}$ to capture these relatively stable relationships induced by the $\mathcal{Z}$.

• *MAV:* To investigate the modeling of contextual temporal information for multivariate attributes MAV, we establish a trainable embedding $C^{\mathrm{mav}}_{\mathrm{emb}} \in \mathbb{R}^{c\times t\times d}$, taking into account "$t$" for inherent temporal correlation differences among various attributes in Fig. 1 left. We also initialize two projections $W^{\mathrm{mav}}_i, W^{\mathrm{mav}}_j \in \mathbb{R}^{d\times d}$, in order to enhance the representational capacity and distinguishability of learnable embeddings. Specifically, $C^{\mathrm{mav}}_{\mathrm{emb}} W^{\mathrm{mav}}_{i,j}$ can be respectively interpreted as the source and target variable during state transfer of MAV (e.g., weather systems variation with sudden temperature drops). Through matrix multiplication guided by Einstein Summation Convention (i.e., dimensional changes "$\mathbb{R}^{c\times t\times d}, \mathbb{R}^{c\times t\times d} \Rightarrow \mathbb{R}^{c\times c\times t}$" using $\otimes$), we obtain $G^{\mathrm{mav}}_{\mathrm{adp}}$:

$$G^{\mathrm{mav}}_{\mathrm{adp}} = \mathrm{Dropout}(\mathrm{Softmax}(\mathrm{Relu}(C^{\mathrm{mav}}_{\mathrm{emb}} W^{\mathrm{mav}}_i \otimes C^{\mathrm{mav}}_{\mathrm{emb}} W^{\mathrm{mav}}_j))) \tag{8}$$

here $\mathrm{Relu}(\cdot)$ ensures the sparsity and reduces the redundant relations for parameters saving; $\mathrm{Softmax}(\cdot)$ normalizes the G_{adp} and maintains robustness; $\mathrm{Dropout}(\cdot)$ avoids embed overfitting during interaction, $G^{\mathrm{mav}}_{\mathrm{adp}} \in \mathbb{R}^{c\times c\times t}$ is time-varying form.

• *MEV:* To model inter-entity correlations, we initialize trainable embeddings $C^{\mathrm{mev}}_{\mathrm{emb}} \in \mathbb{R}^{c\times d}$ for each entity variable, excluding the temporal dimension "t" for $C^{\mathrm{mev}}_{\mathrm{emb}}$ is due to MEV's similarities of temporal correlations in Fig. 1 right. Subsequently, we employ two projections $W^{\mathrm{mev}}_i, W^{\mathrm{mev}}_j \in \mathbb{R}^{d\times d}$ to enhance the representational capacity and differences of entity embeddings for adapting intricate pattern entanglement (e.g., traffic flow is affected by multiple factors and not only at the transportation level). We then construct $G^{\mathrm{mev}}_{\mathrm{adp}}$ using matrix transpose multiplication from global perspective:

$$G^{\mathrm{mev}}_{\mathrm{adp}} = \mathrm{Dropout}(\mathrm{Softmax}(\mathrm{Relu}(C^{\mathrm{mev}}_{\mathrm{emb}} W^{\mathrm{mev}}_i \times C^{\mathrm{mev}}_{\mathrm{emb}} W^{\mathrm{mev}T}_j))) \tag{9}$$

here $G^{\mathrm{mev}}_{\mathrm{adp}} \in \mathbb{R}^{c\times c}$ is an embedding-enhanced self-adaptive graph. C_{emb} makes the quadratic space complexity $O(c^2)$ of G_{adp} to $O(c)$ with input variates in initial stage.

• *MAV-MEV Mixing:* Notably, while mainstream MTSF benchmarks can be categorized into attribute-dominant and entity-dominant types, hybrid scenarios with ambiguous feature representations commonly exist. To address this, we integrate a time-varying graph $G^{\mathrm{mav}}_{\mathrm{adp}} \in \mathbb{R}^{c\times c\times t}$ and a global enhanced graph $G^{\mathrm{mev}}_{\mathrm{adp}} \in \mathbb{R}^{c\times c}$ via adaptive gating fusion, where a dominance coefficient $\alpha_c \in \mathbb{R}^{c\times 1\times 1}$ is learned to indicate feature tendency: $\mathrm{Sigmoid}(\alpha_c) \to 1$ favors attribute features whereas

$\text{Sigmoid}(\alpha_c) \rightarrow 0$ emphasizes entity characteristics. The resultant hybrid graph $G_{\text{adp}}^{\text{mix}} \in \mathbb{R}^{c\times c\times t}$ effectively handles mixed attributes and entities feature scenarios in discussion 6.2, and its graph generation process can be formulated as:

$$G_{\text{adp}}^{\text{mix}} = \text{Sigmoid}(\alpha_c) \cdot G_{\text{adp}}^{\text{mav}} + (1 - \text{Sigmoid}(\alpha_c)) \cdot G_{\text{adp}}^{\text{mev}} \tag{10}$$

here $\text{Sigmoid}(\cdot)$ is used to constrain the gating weights, while α_c will automatically find the optimal "identity" for each time series variable as the model minimizes the prediction loss. $G_{\text{adp}}^{\text{mev}}$ by broadcasting on the time-axis "t" to match $G_{\text{adp}}^{\text{mav}}$.

Channels-Time Graph for Dynamization. Owing to the dynamic nature of VT-ITGraph manifested in temporal evolution, we further augment it with a Channels-Time Graph $G_{\text{CT}} \in \mathbb{R}^{c\times t}$ to self-learning insights into the temporal variations of MTS, meanwhile the learning outcomes of Adaptive Graph from MAV/MEV is aggregated. Thus, we get transferred VT-ITGraph weight $W^{(\text{VT-ITG})}$ in temporal context:

$$P(G_{\text{CT}} \in \{\mathcal{A}\}) = \sum\nolimits_{X\in\{\mathcal{O}\}} P(G_{\text{adp}}, X) P(G_{\text{CT}}|G_{\text{adp}}, X), \tag{11}$$

$$W_{\text{mav}}^{(\text{VT-ITG})} = \sigma_{\text{gelu}}(X^{(\text{ASFF-MLP})} \otimes G_{\text{adp}}^{\text{mav}} \otimes G_{\text{CT}}), \quad \in \mathbb{R}^{b\times c\times t} \tag{12}$$

$$W_{\text{mev/mix}}^{(\text{VT-ITG})} = \sigma_{\text{tanh}}(X^{(\text{ASFF-MLP})} \otimes G_{\text{adp}}^{\text{mev/mix}} \otimes G_{\text{CT}}), \quad \in \mathbb{R}^{b\times t\times t} \tag{13}$$

$$X^{(\text{VT-ITG})} = X^{(\text{ASFF-MLP})} \otimes W_{\text{mav/mev/mix}}^{(\text{VT-ITG})} + X^{(\text{ASFF-MLP})}, \quad \in \mathbb{R}^{b\times c\times t} \tag{14}$$

here $W^{(\text{VT-ITG})}$ in MAV task $\in \mathbb{R}^{b\times c\times t}$, "$c \times t$" indicates diverse attributes temporal trend; In MEV and MAV-MEV mixed task $\in \mathbb{R}^{b\times t\times t}$, "$t \times t$" means global temporal correlations. $\sigma(\cdot)$ denotes activation function (gelu or tanh) to enhance non-linear capability. G_{CT} essentially relies on the pre-learning of G_{adp} and X, but more importantly it establishes a transformation between multivariate correlations to temporal context, which leads to the mining for more explicit or latent temporal patterns.

4.3 Frequency Dilated Group Convolution

Multi-source Signals Fusion. Multi-source feature learning exploits the complementarities between different features, therefore we denominate the original input $X^{(\text{Org})}$, the smoothed input $X^{(\text{Smooth})}$, and the input after multivariate relationship modeling $X^{(\text{VT-ITG})}$ as "multi-source signals". The Fast Fourier Transform (FFT) is applied to these signals to extract their respective real and imaginary frequency components, thereby obtaining a fused representation $H_{\text{fusion}} \in \mathbb{R}^{b\times c\times 2n\times(\lfloor \frac{t}{2} \rfloor+1)}$ by concatenation for complementarities, where n is set to be three types of input.

Patterns Mining and Integration. We introduce stackable and parallel dilated group convolutions for preventing model collapse cased by computational complexity in long-term predictions. Simultaneously, AvgPool has been extensively employed in research on the separation of trend components (Wu et al. 2021; Zhou et al. 2022; Zeng et al. 2023), facilitating the isolation of residual terms and the acquisition of primary temporal patterns from "multi-source signals" in frequency domain:

$$H_{\text{fusion}}^{(l)} = \text{AvgPool}(\text{DG-Conv}^{(l-1)}(\text{Pad}(H_{\text{fusion}}^{(l-1)}))), \quad l = l + 1, \tag{15}$$

$$X^{(\mathrm{FDG})} = \mathrm{Conv2d}(H^{(L)}_{\mathrm{fusion}}) + X^{(\mathrm{Org})} + X^{(\mathrm{VT\text{-}ITG})} + X^{(\mathrm{Smooth})}, \quad l = L \tag{16}$$

here $\mathrm{Pad}(\cdot)$ denotes zero-padding p_d to preserve MTS sequence length invariance, $\mathrm{DG\text{-}Conv}^{(l)}(\cdot)$ signifies causal convolutions with variable grouping, utilizing distinct kernels for each group to capture local temporal patterns. As l increases, the input temporal information "t" is gradually aggregated while the $H^{(l)}_{\mathrm{fusion}}$ is gradually generated by DG-Conv (i.e., dimensional changes $\mathbb{R}^{b\times c\times 2nd_r^l \times t^{(l)}}, t^{(l)} = \lfloor \frac{(\frac{t^{(l-1)}+p_d(d_r-1)-k}{s}+1)}{2} \rfloor$, historical "t" is aggregated through layer-by-layer recursion while the future temporal dimension is generated by receptive fields expansion within dilated factor d_r and $\mathrm{Conv2d}(\cdot)$ adjustment for $\mathbb{R}^{b\times c\times 2nd_r^L \times 1} \rightarrow \mathbb{R}^{b\times c\times t}$, time complexity is related to $O(tlogt)$). Finally, we fuse $H^{(L)}_{\mathrm{fusion}}$ to obtain output $X^{(\mathrm{FDG})} \in \mathbb{R}^{b\times c\times t}$.

4.4 Model Formulation and Forward Calculation

The Frequency Domain Multivariate Block implements the sequential encapsulation of the modules presented in the previous section 4.1, 4.2, 4.3:

$$X^{(\mathrm{ASFF\text{-}MLP})} = \mathrm{ASFF\text{-}MLP}(X), \tag{17}$$

$$W^{(\mathrm{VT\text{-}ITG})} = \mathrm{VT\text{-}ITGraph}(X^{(\mathrm{ASFF\text{-}MLP})}), \tag{18}$$

$$X^{(\mathrm{VT\text{-}ITG})} = X^{(\mathrm{ASFF\text{-}MLP})} \times \sigma(W^{(\mathrm{VT\text{-}ITG})}) + X^{(\mathrm{ASFF\text{-}MLP})}, \tag{19}$$

$$X^{(\mathrm{FDG})} = \mathrm{FDG\text{-}Conv}(X^{(\mathrm{VT\text{-}ITG})}). \tag{20}$$

Where $X^{(\mathrm{ASFF\text{-}MLP})}$, $X^{(\mathrm{VT\text{-}ITG})}$, $X^{(\mathrm{FDG})}$ all $\in \mathcal{R}^{b\times c\times t}$. Finally for calculation, we regard the forward process of VTITG-ASFFN as an end-to-end learning in Fig. 2:

• *History Embedding:* Learn the historical spatio-temporal correlations through $\mathrm{FDMB}(\cdot)$ stacking: $X^{(l)} = \mathrm{FDMB}^{(l)}_{\mathrm{his}}\left(X^{(l-1)}\right), l = 2, ..., L$. Where $X^{(l)}$ all $\in \mathcal{R}^{b\times c\times i}$.

• *Prediction:* Use the $\mathrm{CI\text{-}MLP}(\cdot)$ to predict $Y \in \mathcal{R}^{b\times c\times o}$: $Y = \mathrm{CI\text{-}MLP}(X^{(\mathrm{L})})$.

• *Future Embedding:* Perform the same operation on Y as History Embedding to get the final MTSF result $\hat{X}$: $Y^{(l)} = \mathrm{FDMB}^{(l)}_{\mathrm{pred}}\left(Y^{(l-1)}\right), l = 2, ..., L$. Where $Y^{(l)}$ all $\in \mathcal{R}^{b\times c\times o}$, $\hat{X} = Y^{(L)}$ and $\mathrm{FDMB}^{(l)}_{\mathrm{his}}, \mathrm{FDMB}^{(l)}_{\mathrm{pred}}$ are parameter-independent.

Noted that the objective of the Future-Embedding is to minimize the distance between the predictive distribution and the true distribution due to the label auto-correlation and temporal distribution shift of future time series. Meanwhile we choose $L1$ loss for VTITG-ASFFN's training because of its robust to noises (Han et al. 2024).

Table 1 Datasets description for MAV and MEV tasks (public repositories at MTSF and DSMT except Finance). All MTS are normalized to eliminate variable scale differences.

Datasets	ETT{m1,m2}	DSMT-D1	Weather	Electricity	Traffic	Solar	QPS	Finance
Time Steps	69680	86400	52696	26304	17544	35040	30240	640
Sampling Intervals	15min	1s	10min	1h	1h	15min	1s	1h
Tasks	MAV	MAV	MAV	MEV	MEV	MEV	MEV	MIX
Variables (Channels)	7	7	21	321	862	137	10	80
Scales	small	small	small	medium	large	medium	small	small
Properties	volatility, periodicity	fluctuation	fluctuation	periodicity	periodicity	periodicity	volatility, trend	volatility
Areas	energy	system	weather	electricity	traffic	energy	software	economic
Date Range	[2016.7,2018.6]	[2023.1.1]	[2020.1,2021.1]	[2016.7,2019.7]	[2016.7,2018.7]	[2006.1,2006.12]	[2022.5,2022.6]	[2024.11, 2025.7]

5 Experiments

5.1 Experimental Setup

Datasets. We summarize 8 MTSF datasets from smart city scenarios in real-world in Table 1. The brief background of these MTSF datasets are: **(1) ETT{m1,m2}** contains seven electricity transformer load and oil features. **(2) Electricity** collects the hourly electricity consumption of 321 customers. **(3) Traffic** describes roadway occupancy as measured by different sensors on San Francisco Bay Area freeways. **(4) Weather** includes 21 weather indicators, which are recorded every 10 minutes in 2020. **(5) QPS** dataset records the number of queries for ten software applications during the May-June 2022 period. **(6) DSMT-D1** consists of commands, external stimuli, and telemetry readings of a simulated complex dynamical system in one day. **(7) Solar** This dataset includes 2006 solar power from 137 PV plants in Alabama. Note that in terms of the time series characteristics, "volatility" describes more dramatic variation, while "fluctuation" refers to small variation. "Periodicity", on the other hand, emphasises regularity and repetition. Unlike Time-Series Library (Wu et al. 2021, 2022), our "custom" dataloader uses a non-overlapping strategy to avoid valid/test sets take historical inputs from the end of the preorder set, so the actual sample size may be slightly different. Meanwhile, ETTm1 still follows the previous "ETT" settings while ETTm2 use our "custom" dataloader to reflect certain intra-domain differences.

Baselines. We select 9 SoTA MTSF models in related work to enhance comparability: (1) **Transformer-based:** iTransformer (Liu et al. 2024), PatchTST (Nie et al. 2023); (2) **MLP-based:** TimeMixer (Wang et al. 2024), DLinear (Zeng et al. 2023), FilterNet (Yi et al. 2024a); (3) **TCN-based:** ModernTCN (Luo et al. 2024), TimesNet (Wu et al. 2022); (4) **Graph-based:** MSGNet (Cai et al. 2024); (5) **LLM-based:** Time-LLM (Bert backbone) (Jin et al. 2024b). In addition, we also introduce RNN-based LSTM, Tree-based GBRT in channel-independent form and statistical model ARIMA.

Settings. All models were executed on NVIDIA GeForce RTX 4090 GPUs. The packages used included Python 3.11.5, NumPy 1.26.4 and PyTorch 2.1.2+cu121. Training experiments were configured utilizing the $L1$ loss and ADAM optimizer for unification. We performed multi-step forecasting at horizons {96,192,336,720} from short to long-term within the same 96 length of look-back windows for comparison fairness. The maximum early stopping epoch is 42 and learning rate is $1e^{-4}$ within 32 batch size. Note that for Time-LLM we adjusted its batch size and text truncation length to fit the max memory 24GB of GPU. The default FDMB($\cdot$) blocks are set to be 2 while embedding dim d of $C_{\text{emb}} = 10$, dilated factor $d_r = 2$, zero-padding size $p_d = 2$, convolution kernel size $k = 3$, stride $s = 1$, stacked layers $L = \lfloor \log_{d_r} \frac{t}{2n} \rfloor$. We use Mean Absolute Error (MAE=$\frac{1}{c \times t} \sum_c \sum_t |Y_{\text{pred}} - Y_{\text{true}}|$) and Mean Squared Error (MSE=$\frac{1}{c \times t} \sum_c \sum_t |Y_{\text{pred}} - Y_{\text{true}}|^2$) as metrics. We did not search excessively for hyperparameters and learning schedule to reflect the inherent baseline differences.

5.2 Experimental Results and Hyperparameters Analysis

• **Comparison Results:** As results shown in Table 2. (1) For MAV task, the models oriented to temporal patterns are outstanding. i.e., TimeMixer and DLinear based

Table 2 Comparison in MAV and MEV tasks (Average results of forecasting horizons {96,192,336,720}). **Red**/**Blue**/**Orange** indicate **1st**/**2nd**/**3rd**. The [train, valid, test] ratio is 0.6:0.2:0.2 for ETTm1 and 0.7:0.1:0.2 for other MTSF datasets.

Models	VTITG-ASFFN		Time-LLM		iTransformer		TimeMixer		ModernTCN		FilterNet		MSGNet		TimesNet		PatchTST		DLinear		GBRT		LSTM		ARIMA	
Metric	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
ETTm1	**0.388**	**0.382**	0.403	**0.388**	0.410	0.407	**0.388**	0.394	0.406	0.393	**0.401**	0.392	0.411	0.398	0.406	0.399	**0.400**	**0.387**	0.407	0.403	0.426	0.441	0.572	0.711	0.698	1.039
ETTm2	**0.264**	**0.161**	**0.272**	0.169	0.286	0.181	**0.270**	**0.167**	0.274	0.175	**0.270**	**0.168**	0.282	0.184	0.275	0.172	0.290	0.172	0.283	0.170	0.289	0.187	0.470	0.414	0.308	0.181
DSMT-D1	**0.157**	**0.092**	0.181	0.120	0.186	0.113	**0.174**	0.119	**0.170**	0.121	0.175	0.120	0.185	0.132	0.182	**0.104**	0.204	0.120	**0.170**	**0.097**	0.215	0.127	0.431	0.310	0.196	0.114
Weather	**0.245**	**0.213**	0.276	0.263	0.278	0.258	**0.262**	0.242	**0.257**	**0.233**	0.272	0.245	0.278	0.249	0.264	**0.224**	0.272	0.235	0.317	0.265	0.291	0.257	0.358	0.335	0.303	0.255
MAV-Average	**0.264**	**0.212**	0.283	0.235	0.290	0.240	**0.273**	0.231	**0.277**	0.230	0.280	0.231	0.289	0.241	0.282	**0.225**	0.292	**0.229**	0.294	0.234	0.305	0.253	0.458	0.443	0.376	0.397
QPS	**0.188**	**0.096**	0.404	0.447	0.266	0.253	0.284	0.232	**0.171**	**0.115**	0.277	0.227	**0.169**	**0.142**	0.326	0.279	0.278	0.154	0.271	0.174	0.316	0.270	0.421	0.365	0.572	0.587
Traffic	**0.283**	0.495	0.311	0.526	**0.296**	**0.459**	0.298	**0.485**	0.335	0.497	0.309	**0.486**	0.342	0.657	0.336	0.620	0.331	0.594	0.383	0.625	**0.288**	0.560	0.444	0.885	0.826	1.477
Electricity	**0.259**	**0.165**	0.289	0.210	**0.270**	**0.178**	**0.273**	**0.182**	0.289	0.188	0.281	0.204	0.300	0.194	0.295	0.193	0.290	0.187	0.300	0.212	0.278	0.192	0.413	0.357	0.771	0.943
SoLar	**0.213**	**0.208**	0.241	0.223	0.242	0.221	0.249	0.251	0.241	0.218	0.242	0.227	0.257	0.245	0.263	0.259	**0.221**	0.270	0.251	0.291	**0.224**	**0.191**	0.244	**0.213**	0.762	1.068
MEV-Average	**0.236**	**0.241**	0.311	0.351	0.268	**0.278**	0.276	0.288	**0.259**	**0.255**	0.277	0.286	**0.267**	0.310	0.305	0.337	0.280	0.301	0.301	0.325	0.276	0.303	0.426	0.455	0.732	1.019

on decomposition and mixing; ModernTCN and TimesNet for patterns extraction; FilterNet for frequency filtering process; Time-LLM and PatchTST using the patch window. However, VTITG-ASFFN uses Inter-Temporal Graph to transfer the multivariate correlations into the temporal context to synergize with frequency components fusion learning, exhibiting the overwhelming performance. (2) For MEV task, VTITG-ASFFN almost achieves the most superior performance. Further observations reveal that Time-LLM struggles to predict in periodic and volatile datasets (e.g., QPS) within limited input time window, increasing look-back window might help but would lead to unfairness in inconsistent size of test set (Qiu et al. 2024). Although the multi-scale graph of MSGNet is suitable for small-scale entities (QPS), it fails in the case of large-scale entities (Traffic). Despite TimeMixer performs de-entanglement, it still fails to conduct effective pattern decomposition in the scenarios with lack of solar energy (Solar), whereas VTITG-ASFFN is compatible with all the above scenarios, demonstrating its multi-scenario scalability. The overall comparison of average performance at 8 real-world datasets underscores the effectiveness of VTITG-ASFFN. Specifically, it realizes SoTA improvement of 3.54%, 5.53% (MAE, MSE) for MAV tasks and 6.61%, 5.08% (MAE, MSE) for MEV tasks. However, VTITG-ASFFN still has room for enhancing dynamic graph quality since its MSE slightly inferior to iTransformer in large-scale variable scenarios (Traffic). Meanwhile LSTM, GBRT, ARIMA are also comparable in some clear periodicity and trend scenarios, therefore machine learning, simple deep learning and traditional methods should not be dismissed.

• **Hyperparameters Analysis:** Frequency Domain Multivariate Blocks significantly influence the performance of VTITG-ASFFN. To illustrate this, we analyze the MAE performance of FDMB(·) with varying numbers of stacked layers across different MAV/MEV datasets, focusing on History-Embedding learning in Table 3, which visually demonstrates that choosing lesser stacked layers ($1 \sim 2$) yields good performance in MAV for its fewer channels, and more layers have the possibility in leading more worse performance because of over-parameterization; Since MEV typically has more

Table 3 Hyperparameter analysis of stacked FDMB layers on MAV and MEV datasets.

Dataset	Type	Metric	Number of Stacked Layers				
			1	2	3	4	5
Weather	MAV	MAE	**0.178**	0.187	0.190	0.193	0.186
Traffic	MEV	MAE	0.289	0.275	0.271	**0.269**	0.272

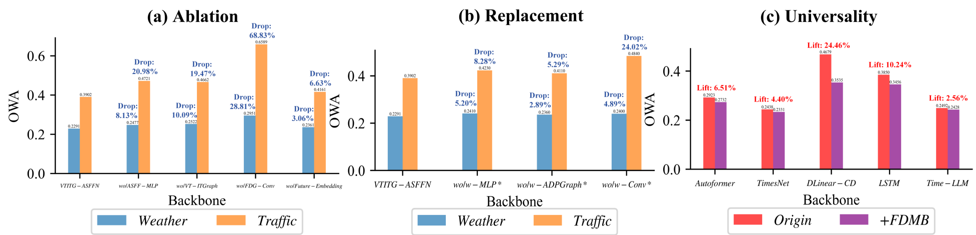

Fig. 4 Ablation and Universal Component Study. We use Overall Weighted Average (OWA) to indicate average variation of MAE/MSE in {96,192,336,720} horizons forecasting.

variables than MAV (refer to scales in Table 1), appropriate stacked layers ($2 \sim 4$) is instrumental better capturing intricate multivariate dependencies.

6 Discussion

We validated the domain impacts of VTITG-ASFFN for MTSF challenges of section 1: **(RQ1)** Are the core modules effective? **(RQ2)** Can it suitable for MAV-MEV mixed MTSF scenarios? **(RQ3)** How interpretable in capturing dynamic dependencies and MAV/MEV properties? **(RQ4)** Does learning patterns show local-global collaboration and it is more advantageous for prediction consistency than other frequency domain models? **(RQ5)** How does adaptive smoothing manifest in real-world scenarios? **(RQ6)** Can it handle ultra-long MTSF within efficient operational efficiency?

6.1 Ablation and Universal Component Study (RQ1)

We conduct ablations in Weather and Traffic datasets on behalf of MAV/MEV scenarios. The notations in Fig. 4: **"wo/(ASFF-MLP/VT-ITGraph/FDG-Conv/Future-Embedding)"** signify these components outputs are zeroed out to isolate contributions, **"wo/w-(MLP/ADPGraph/Conv)*"** indicate replacing ASFF-MLP/VT-ITGraph/FDG-Conv with MLP/Adaptive Graph/Convolution.

From Fig. 4 we find: (1) Upon the removal of all the proposed contribution modules, we consistently observe a significant decrement in the performance of VTITG-ASFFN, as evidenced by notable increase of OWA in Fig. 4 (a). And VTITG-ASFFN also outperforms more basic MLP/Graph/Conv-based alternatives in replacement, which can be confirmed by the relative OWA increase in Fig. 4 (b). (2) The removal of frequency-based modules (i.e., ASFF-MLP, FDG-Conv) severely impairs the capacity of VTITG-ASFFN to uncover latent temporal patterns within the frequency domain. (3) The removal of Variable-Tailored Inter-Temporal Graph results in the loss of "dynamic spatial-temporal dependencies", leading to a decline in prediction. (4) The Future-Embedding is generally effective because it further exploits the latent multivariate dependencies and temporal information in the forecasting results of VTITG-ASFFN. It can be regarded as a complement to the History-Embedding by autonomously learning future information rather than stacking FDMB($\cdot$) based on the historical time window. However, it may no longer valid in the ultra-long term in discussion 6.6. (5) Without loss of generality, we choose Transformer-based (Autoformer),

Table 4 MAV-MEV mixed comparisons in the synthetic "Finance" dataset (input-12-predict-12).

Models	Ours (MIX)		Ours (MAV)		Ours (MEV)		iTransformer		TimeMixer		ModernTCN		Time-LLM		MSGNet	
Metric	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
Finance	**0.346**	**0.400**	0.476	0.703	0.355	0.444	<u>0.347</u>	<u>0.404</u>	0.350	0.413	0.367	0.462	0.355	0.419	0.422	0.546

TCN-based (TimesNet), MLP-based (DLinear-CD), RNN-based (LSTM), LLM-based (Time-LLM) as backbones. From Electricity dataset plugin shown in Fig. 4 (c), we find FDMB(·) is applicable to mainstream deep learning networks, exhibits excellent universality, and has the most prominent improvement on MLP-based, facilitating the lightweight models for accurate forecasting. Although the effect of the FDMB(·) diminishes as the backbone capability increases, it still provides substantial benefits.

6.2 MAV-MEV Mixing Study (RQ2 for *Challenge I*)

We selected 20 representative blue-chip stocks from companies listed on the Shenzhen Stock Exchange that represent manufacturing sector and real economy of China. For each stock, we extracted four price behavior attributes: open, close, high, and low. As for the VTITG-ASFFN model, we considered three design variations for the VT-ITGraph: one tailored for MAV, one for MEV, and a hybrid MAV-MEV approach.

From Table 4 we find that the VTITG-ASFFN variant utilizing the hybrid MAV-MEV design achieves the best performance on quantitative metrics, significantly outperforming the single-logic variants that use only the "Multi-Attribute" or "Multi-Entity" approach. This finding indicates that in scenarios hard to pre-define variables as either "Multi-Attribute" or "Multi-Entity", employing adaptive gating mechanism to fuse different variable-tailored graphs provides enhanced generalization ability.

6.3 Interpretability Study (RQ3 for *Challenge I, II*)

With the VT-ITGraph as the centerpiece, we demonstrate the interpretability of VTITG-ASFFN for "dynamic spatial-temporal dependencies" and multivariate natures capture for "Multi-Attribute" and "Multi-Entity" time series variables.

• **Dynamic Spatial-Temporal Dependencies.** Within the Traffic dataset, we extracted the self-adaptive graphs first. Starting from Fig. 5 left to right, upon examining localized areas, grids with deeper red hues indicate prominently associated sensor

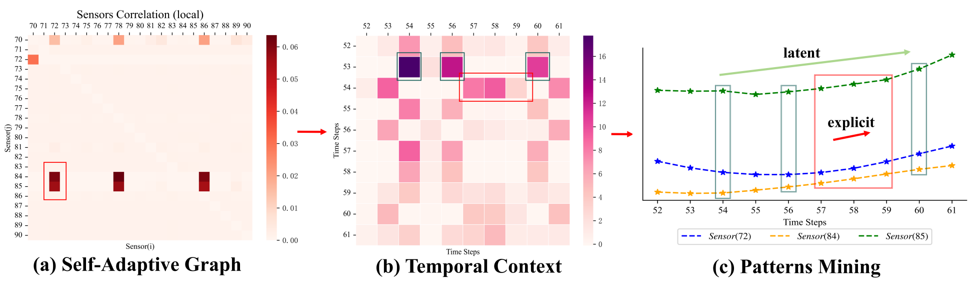

Fig. 5 The capture of explicit and latent "dynamic spatial-temporal dependencies" in Traffic. Left: (a) self-adaptive graph; Middle: (b) temporal context; Right: (c) patterns mining.

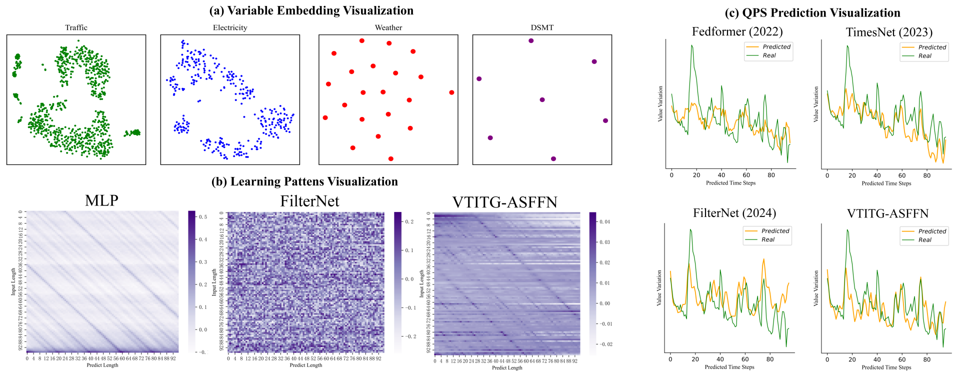

Fig. 6 (a): Variable Embedding Visualization; (b): Learning Patterns Visualization; (c) Real prediction consistency compared to frequency domain-based MTSF models.

pairs, for instance, a notable correlation exists among the id of sensor 72, 84, 85. VTITGraph then transfers this correlation into the temporal context, and we select two patterns of significant temporal correlations in the red/green boxes. Where the red box responds to collaborative changes in sensor 72, 84, 85 under time periods 57 to 59, and the green box reflects a latent upward trend among time points 54, 56, and 60. This phenomenon reflects diverse dependencies (explicit or latent) among multivariate time series, embodying the concept of "dynamic spatial-temporal dependencies".

• **MAV and MEV Variable Embedding Visualization.** We visualize the variable embeddings by t-SNE in 2D form. From Fig. 6 (a), which demonstrates that the entity variables in Traffic and Electricity datasets reflect spatial clustering, whereas the attribute variables in Weather and DSMT are independently distributed, reflecting the conceptual divide, which is consistent with the inherent physical attributes describing the weather state and the complex system. We validate the effectiveness of the variable-tailored adaptive graphs in capturing the intrinsic nature of MAV and MEV.

6.4 Learning Pattern and Prediction (RQ4 for *Challenge III*)

We choose the linear mapping weights of MLP, FilterNet and VTITG-ASFFN in QPS dataset, while using 96 input length for 96 future horizons forecasting. We leverage weight visualization to reveal the underlying learning patterns of VTITG-ASFFN.

From Fig. 6 (b): (1) Upon observing the MLP, the insufficient leverage of all pattern connections reveals the information utilization bottleneck in the point-wise mapping approach of MLP, which prevents it from handling the global dependencies within time series. (2) In analyzing the FilterNet, we find the learnable frequency filter explicits global temporal patterns. However, it does not exhibit a diagonal relationship, reflecting the neglect of local components. (3) VTITG-ASFFN enriches the global periodic patterns and dependencies through frequency fusion. Concurrently, it maintains a certain level of reliance on diagonal patterns, emphasizing its focus on local components exhibiting compact energy distribution through the embedding learning of CI-MLP. Furthermore, the collaborative local-global learning results in the superior prediction consistency of VTITG-ASFFN compared to other frequency-based in Fig. 6 (c).

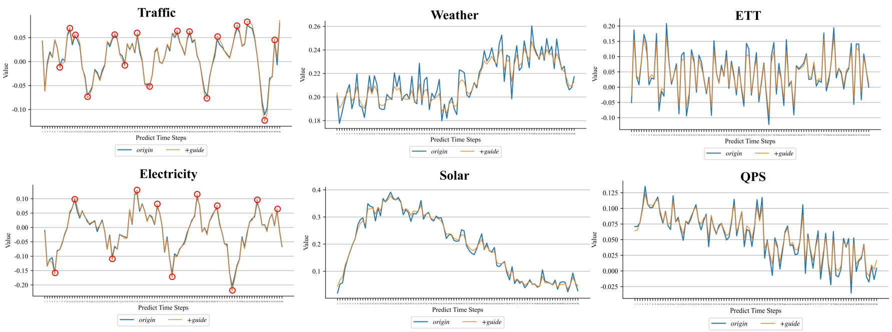

Fig. 7 Adaptive-Smoothing by VTITG-ASFFN in [periodicity, fluctuation, volatility] MTSF scenarios from left to right, where "+guide" denotes using adaptive mask. The variable channels of MTS are normalized and averaged to represent the overall trend.

6.5 Adaptive-Smooth Study (RQ5 for *Challenge III*)

In Fig. 7, we demonstrate the deployed application of adaptive-smooth for diverse multivariate time series stationarization. We present the scenarios in order of periodicity, fluctuation, volatility: (1) For the Traffic and Electricity datasets, VTITG-ASFFN stabilizes and amplifies peak information (see red circles in Fig. 7), facilitating the prediction of cross-variable abrupt events, such as a sudden increase in traffic congestion due to regional accidents or unexpected surges in electricity demand indicating anomalous states, which are crucial for proactive management. (2) For the Weather and Solar datasets, which exhibit complex meteorological patterns and uncertainties, VTITG-ASFFN mitigates the fluctuation of prediction and clearly captures the evolution of trends, enabling more reliable forecasting. (3) For the ETT and QPS datasets with frequent and intense magnitude variation in highly dynamic systems, VTITG-ASFFN reduces peaks to mitigate noise impact, while maintaining consistent volatility of trends despite rapid changes in the original MTS. In summary, adaptive-smooth could provide targeted guidance for various real-world scenarios.

6.6 Ultra-Long Forecasting and Operational Efficiency (RQ6)

Existing MTSF studies rarely consider the ultra-long time series inference. We choose million-scale datasets DSMT with 1048574 time points and 7 variables, and select the latest framework including Transformer, Linear, TCN, Graph and LLM based for comparison (VTITG-ASFFN* indicates without Future-Embedding). However, VTITG-ASFFN's parameters grow rapidly with prediction length, which is relevant to the end-to-end architecture since we also do FDMB($\cdot$) embedding learning for the output MTS. As the distribution drift (Han et al. 2024) and heterogeneity of ultra-long term MTS become more significant, which will lead to unreliability of latent temporal patterns mined by Future-Embedding (ineffective results in 2th row of Table 5). Therefore we remove the Future-Embedding to avoid error accumulation and the additional cost of ultra-long forecasting. As shown in Table 5, VTITG-ASFFN achieves superior inference performance, with SoTA improvement of (4.86%, 8.24%) in (MAE, MSE) and (27.97%, 33.33%) in (MB, s/iter), confirming its computing capability. In

Table 5 Ultra-Long MTSF and Operational Efficiency in DSMT (512 input length). VTITG-ASFFN is efficient due to low embed dim VT-ITGraph and parallel FDG-Conv.

Horizons	512		1024		2048		3072		Average		Memory.avg	Speed.avg
Metric	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	(MB)	(s/iter)
VTITG-ASFFN*	**0.199**	**0.125**	**0.258**	**0.191**	**0.307**	**0.249**	**0.332**	**0.281**	**0.274**	**0.211**	**886**	**0.0024**
VTITG-ASFFN	**0.202**	**0.129**	**0.259**	**0.191**	**0.307**	**0.249**	**0.334**	**0.282**	**0.276**	**0.213**	3090	**0.0067**
iTransformer	0.217	0.147	0.277	0.220	0.338	0.301	0.370	0.342	0.301	0.253	**1582**	**0.0036**
TimeMixer	0.216	0.142	0.282	**0.216**	**0.335**	**0.279**	0.357	**0.311**	0.298	0.237	2340	0.0110
ModernTCN	0.218	0.154	0.283	0.239	0.345	0.321	0.380	0.368	0.307	0.271	**1230**	0.0074
MSGNet	0.230	0.163	0.295	0.249	0.359	0.338	0.400	0.400	0.321	0.288	2952	0.0498
Time-LLM	**0.206**	**0.134**	**0.267**	**0.203**	**0.323**	**0.274**	**0.356**	0.314	**0.288**	**0.231**	16036	0.2135

addition, ultra-long forecasting implies that we should judiciously utilize the FDMB$(\cdot)$ for balance between prediction performance and computational consumption.

7 Conclusion and Future Work

In this research, we introduce VTITG-ASFFN for multivariate time series forecasting. VTITG-ASFFN undertakes adaptive smoothing of time series as a prelude to frequency domain analysis. Subsequently, it amalgamates frequency domain amplitude and phase information for local-global learning. Moreover, Variable-Tailored Inter-Temporal Graph offers tailored adaptive multivariate correlation techniques for both "Multi-Attribute" and "Multi-Entity" forecasting tasks and transfer the correlations to temporal context, facilitating the capture of "dynamic spatial-temporal dependencies". Comprehensive evaluations on eight real-world datasets demonstrate its superiority for MTSF applications. For future studies we could explore the overall performance of generalized time series analysis to fully measure model generalizability.
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References

Bai L, Yao L, Li C, et al (2020) Adaptive graph convolutional recurrent network for traffic forecasting. Advances in neural information processing systems 33:17804–17815

Cai W, Liang Y, Liu X, et al (2024) Msgnet: Learning multi-scale inter-series correlations for multivariate time series forecasting. In: AAAI Conference on Artificial Intelligence

Casals J, Jerez M, Sotoca S (2000) Exact smoothing for stationary and non-stationary time series. International Journal of Forecasting 16(1):59–69

Chen T, Guestrin C (2016) Xgboost: A scalable tree boosting system. In: acm sigkdd international conference on knowledge discovery and data mining, pp 785–794

Fan W, et al (2024) Deep frequency derivative learning for non-stationary time series forecasting. In: International Joint Conference on Artificial Intelligence

Han L, Du B, Sun L, et al (2021) Dynamic and multi-faceted spatio-temporal deep learning for traffic speed forecasting. In: ACM SIGKDD conference on knowledge discovery & data mining, pp 547–555

Han L, Ye HJ, Zhan DC (2024) The capacity and robustness trade-off: Revisiting the channel independent strategy for multivariate time series forecasting. IEEE Transactions on Knowledge and Data Engineering

Huang Q, Shen L, Zhang R, et al (2023) Crossgnn: Confronting noisy multivariate time series via cross interaction refinement. In: Advances in Neural Information Processing Systems

Jiang R, Wang Z, Yong J, et al (2023) Spatio-temporal meta-graph learning for traffic forecasting. In: AAAI conference on artificial intelligence, pp 8078–8086

Jin M, Koh HY, Wen Q, et al (2024a) A survey on graph neural networks for time series: Forecasting, classification, imputation, and anomaly detection. IEEE Transactions on Pattern Analysis and Machine Intelligence

Jin M, et al (2024b) Time-LLM: Time series forecasting by reprogramming large language models. In: International Conference on Learning Representations

Liu Y, Hu T, Zhang H, et al (2024) itransformer: Inverted transformers are effective for time series forecasting. In: International Conference on Learning Representations

Loweimi E, Yue Z, Bell P, et al (2023) Multi-stream acoustic modelling using raw real and imaginary parts of the fourier transform. IEEE/ACM Transactions on Audio, Speech, and Language Processing

Luo D, et al (2024) ModernTCN: A modern pure convolution structure for general time series analysis. In: International Conference on Learning Representations

Nie Y, Nguyen NH, Sinthong P, et al (2023) A time series is worth 64 words: Long-term forecasting with transformers. In: International Conference on Learning Representations

Papastefanopoulos V, Linardatos P, Panagiotakopoulos T, et al (2023) Multivariate time-series forecasting: A review of deep learning methods in internet of things applications to smart cities. Smart Cities

Pei SC, Huang SG (2012) Stft with adaptive window width based on the chirp rate. IEEE Transactions on Signal Processing

Qiu X, Hu J, Zhou L, et al (2024) Tfb: Towards comprehensive and fair benchmarking of time series forecasting methods. In: VLDB Endowment

Shao Z, Zhang Z, Wang F, et al (2022) Spatial-temporal identity: A simple yet effective baseline for multivariate time series forecasting. In: ACM International Conference on Information & Knowledge Management

Shumway RH, Stoffer DS, Shumway RH, et al (2017) Arima models. Time series analysis and its applications: with R examples pp 75–163

Sun L, et al (2022) Tess: multivariate sensor time series prediction for building sustainable smart cities. ACM Transactions on Sensor Networks

Torres JF, et al (2018) A scalable approach based on deep learning for big data time series forecasting. Integrated Computer-Aided Engineering

Wang S, Wu H, Shi X, et al (2024) Timemixer: Decomposable multiscale mixing for time series forecasting. In: International Conference on Learning Representations

Wu H, Xu J, Wang J, et al (2021) Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. In: Advances in Neural Information Processing Systems

Wu H, Hu T, Liu Y, et al (2022) Timesnet: Temporal 2d-variation modeling for general time series analysis. In: International Conference on Learning Representations

Wu Z, Pan S, Long G, et al (2020) Connecting the dots: Multivariate time series forecasting with graph neural networks. In: ACM SIGKDD international conference on knowledge discovery & data mining, pp 753–763

Wu Z, et al (2019) Graph wavenet for deep spatial-temporal graph modeling. In: International Joint Conference on Artificial Intelligence

Yi K, Fei J, Zhang Q, et al (2024a) Filternet: Harnessing frequency filters for time series forecasting. In: Advances in Neural Information Processing Systems

Yi P, Huang F, Peng J, et al (2024b) Dynamic spatial-temporal embedding via neural conditional random field for multivariate time series forecasting. ACM Trans Spatial Algorithms Syst

Zeng A, Chen M, Zhang L, et al (2023) Are transformers effective for time series forecasting? In: AAAI Conference on Artificial Intelligence

Zhang T, et al (2022) Less is more: Fast multivariate time series forecasting with light sampling-oriented mlp structures. arXiv preprint arXiv:220701186

Zhou T, Ma Z, Wen Q, et al (2022) Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In: International Conference on Machine Learning, PMLR

Zhuang W, Fan J, Fang J, et al (2024) Rethinking general time series analysis from a frequency domain perspective. Knowledge-Based Systems